

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (April – 2025)

M. Sc. Digital Forensics and Information Security

Semester – II

Subject Code: CTMSDFIS24 SII P1

Date: 24/04/2025

Subject Name: Network Security & Forensic

Time: 10:30AM-1:30PM

Total Marks: 100

Instructions:

1. Write down each question on a separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

			Marks
Q.1		Attempt any three.	
✓ (a)	✓	Explain the OSI and TCP/IP models in detail. Compare their layers and functionalities.	08
✓ (b)	✓	Describe different perimeter devices like IDS, IPS, and Firewalls. How do they contribute to network security?	08
✓ (c)	✓	Explain the working of common network devices like switches, routers, and hubs with their functions in network topology	08
✓ (d)	✓	Discuss different types of cyber-attacks such as ARP poisoning, MAC flooding, and DoS/DDoS.	08
Q.2		Attempt any three.	
✓ (a)	✓	Explain the penetration testing life cycle. Describe each phase with real-world examples.	08
✓ (b)	✓	Discuss the types of scanning techniques used in penetration testing.	08
(c)		What is SNMP enumeration and how is it used in penetration testing?	08
✓ (d)	✓	Explain password cracking techniques and countermeasures.	08
Q.3		Attempt any three.	
✓ (a)	✓	Describe symmetric and asymmetric encryption with examples.	08
(b)		Explain the concept of digital signatures and their verification process.	08
(c)		Write notes on hash functions and message digests. How are they used in security?	08
✓ (d)	✓	What is RSA algorithm. Given two prime numbers, $p=61$ and $q=53$, calculate the RSA public and private keys. $e=2$	08
Q.4		Attempt any two.	
(a)		Describe the 802.11 protocol and its security features.	07
✓ (b)	✓	What are WLAN attacks and how can they be prevented?	07
✓ (c)	✓	Explain DHCP and DNS Servers in the details.	07

Q.5		Attempt any two.	
	(a)	What is digital evidence? Explain its types and importance in network forensics.	07
	(b)	Describe the process of capturing and analyzing network traffic for forensic investigation.	07
	(c)	Explain the key principles of the OSCAR methodology in digital forensic and its implementation during investigation.	07

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NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (April – 2025)

M.Sc. Digital Forensics and Information Security
Semester – II

Subject Code: CTMSDFIS SII - P2

Date: 25/04/2025

Subject Name: Web Application Security

Time: 10.30 AM - 01.30 PM

Total Marks: 100

Instructions:

1. Write down each question on a separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

- Q.1** Attempt any three. Marks
- (a) Explain NMAP and Its Scanning Techniques. 08
- (b) What is Threat Modeling? Explain any two Models. 08 *stride & dward.*
- (c) What is Cross Site Scripting (XSS)? Explain how it can affect a website. 08
- (d) What do you understand by Sensitive Data Exposure? Give two examples of exposed data? 08
- Q.2** Attempt any three.
- (a) Explain the lifecycle of a Vulnerability Assessment. Illustrate each phase with suitable examples? 08
- (b) What is a Web Service? Explain the difference between Web Service and Website? 08
- (c) What is IDOR? Give a simple example. 08
- (d) Write a note on HTTP and its methods 08
- Q.3** Attempt any three.
- (a) What are CVE and CWE? How are they useful in the context of vulnerability management? Provide relevant examples. 08
- (b) What is CMS? and Why CMS Security is important. 08
- (c) Explain File Upload Vulnerability. 08
- (d) What is Insecure Captcha? How can attackers bypass a weak CAPTCHA system? 08
- Q.4** Attempt any two.
- (a) How do you handle a data breach during an incident? 07
- (b) Explain the role of Cookies and Sessions in Web Applications? 07
- (c) What is SQL Injection? Write a simple example to show how it works? 07

Q.5 **Attempt any two.**

- ✓(a) What is Docker? Write a note on Advantages of docker **07**
- ✓(b) What is Security Misconfiguration? Mention two common examples? **07**
- (c) What are weak authentication tokens? Why are they risky in web applications? **07**

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NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (April – 2025)
M.Sc. Digital Forensics & Information Security
Semester – II

Subject Code: CTMSDFIS24 SII P3

Date: 28/04/2025

Subject Name: Malware Analysis & Forensic

Time: 10:30am To 01:30pm

Total Marks: 100

Instructions:

1. Write down each question on a separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

		Marks
Q.1	Attempt any three.	
(a)	Explain the Process Explorer tool in Detail.	08
(b)	What Volatility Framework? Explain any 5 command of it.	08
(c)	What is the use of Sand Box in Malware Analysis. Explain one Sand box.	08
(d)	Discuss the role of Dynamic Analysis in Malware.	08
Q.2	Attempt any three.	
(a)	What is Malware? Explain any 4 Malware with example.	08
(b)	What is Register? Explain General Purpose Register in Details.	08
(c)	Explain the structure of PE files with Diagram, focusing on the Importance of headers and sections in malware analysis.	08
(d)	What is DLL? Explain its Types and Role in Malware Analysis.	08
Q.3	Attempt any three.	
(a)	What is Yara rule? How to Write it? Explain it one example.	08
(b)	Explain the purpose and basic usage of the reverse engineering tool IDA Free.	08
(c)	Discuss the Lab setup for Malware Analysis in Virtual Machine.	08
(d)	Explain various Flags in x86 CPU architecture.	08
Q.4	Attempt any two.	
(a)	What are the artefacts you can cover from Linux System.	07
(b)	Explain how Dependency Walker can be used to analyse .dll files.	07
(c)	What is DLL Injection? Explain DLL Injection Methods.	07
Q.5	Attempt any two.	
(a)	Discuss any Case Study of Malware Attack.	07
(b)	What is a Debugger, and Explain its various types?	07
(c)	Explain Hooking Injection in details	07

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NATIONAL FORENSIC SCIENCES UNIVERSITY**M.Sc. (DFIS) (Semester II)****Semester End Examination –April- 2025****Subject Code: CTMSDFIS24 SII P4****Date: 29.04.2025****Subject Name: Artificial Intelligence****Time: 10:30 AM to 1:30 PM****Total Marks: 100****Instructions:**

1. Write down each question on a separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

Marks**Q.1****Attempt any three.****08**

- (a) Write a Python Script/Commands for following:

create a 1D array of numbers from 0 to 5 using NumPy, ii) reshape it into a 3*2 array, iii) create a DataFrame from the reshaped array using Pandas, iv) Plot a bar graph of the DataFrame using Matplotlib. v) Save the figure.

OR

Given the data of a laboratory having 5 systems for a month (of 4 weeks), failed Login Attempts, [20,20,24,32,28], write a Python script and User defined function Print_Attacks to calculate all the statistics (mean, mode, median, standard deviation, variance) for the lab per week. Write the results in an excel file.

- (b) Consider the example below where the mass, y (grams) of a chemical is related to the time, x (seconds), for which the chemical reaction has been taking place according to the table:

08

Time, x (seconds) 5 7 12 16 20

Mass, y (grams) 40 120 180 210 240

Find the equation of the regression line. Using resultant regression line predict the data points.

- (c) List out and briefly explain types of Clustering. Discuss the working of K-Means Clustering algorithm.

08

- (d) Bring out the difference between following:

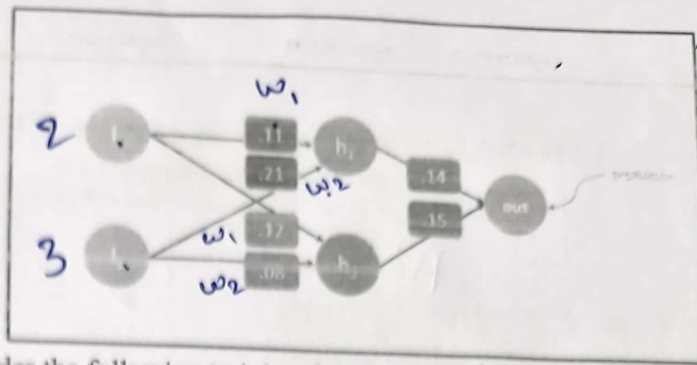
08

- 1) Classification vs Prediction
- 2) Machine Learning vs Deep Learning

Q.2**Attempt any three.**

- (a) Explain in brief the different types of layers in an ANN? Use perceptron algorithm and calculate value for hidden layer neurons (h_1 , h_2). Consider inputs=[2, 3] and actual output=[1].

08



(b) Consider the following training data for the Naive Bayes Classifier

	Type of family structure	Age group	Income status	Will they buy a car?
1	Nuclear	Young	Low	Yes
2	Extended	Old	Low	No
3	Childless	Middle-aged	Low	No
4	Childless	Young	Medium	Yes
5	Single Parent	Middle-aged	Medium	Yes
6	Childless	Young	Low	No
7	Nuclear	Old	High	Yes
8	Nuclear	Middle-aged	Medium	Yes
9	Extended	Middle-aged	High	Yes
10	Single Parent	Old	Low	No

Given test inputs – {Single Parent, Young, and Low}, compute the probability of buying a car.

(c) Describe the types of data quality issues possible in real time dataset. Discuss different steps of Data Pre-Processing.

(d) Write a note on following:

(I) Overfitting (II) Perceptron (III) Image Segmentation (IV) Labelled vs Unlabeled Data

Q.3

Attempt any three.

(a) The following confusion matrix shows data for a Binary-Class problem where classification model predicts whether a person loves Facebook or Instagram. Calculate Accuracy, TN rate, Precision, Recall, Error Rate, and FP Rate.

		Predicted	
		Facebook	Instagram
	Actual		
	Facebook	66	9
	Instagram	22	31

TP FN
FP TN

TP = 66

TN =

(b) Explain the working of a Convolutional Neural Network (CNN), describing its components and how it is used for image classification. Illustrate the process with an example.

(c) Explain the key steps involved in the Natural Language Processing (NLP) pipeline, providing examples of their application.

Sup / unsup
↓

- ✓(d) Discuss in detail the various types of Machine learning models.

08

Q.4

Both the questions are compulsory

- ✓(a) List out and discuss different static as well as dynamic features, which can be used for malware analysis using Machine Learning
- ✓(b) What is activation function? Explain different types of activation function in detail?

07

07

14

Q.5

Both the questions are compulsory

- ✓(a) Define object tracking. List out and discuss types of object tracking. Explain object tracking algorithm with an example
- ✓(b) Explain the architecture of LSTM with help of Diagram. What is significance of different types of gates in the working of LSTM?

07

07

14

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